

KAIYI ZHANG

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SUMMARY

Computer science Ph.D. expert in Optimal Transport and Machine Learning. Senior Data Scientist at Walmart Global Tech delivering end-to-end ML systems with measurable business impact. Experience across generative AI, recommendation systems, LLM post-training, RLHF, generative recommender, retrieval/reranker models, transformers, and GNNs.

Tech Stack: Python, PyTorch, TensorFlow, Hugging Face, Spark, BigQuery, Airflow, Vespa, Docker, GKE/K8s, Databricks, SQL, Scala

PROFESSIONAL EXPERIENCE

Walmart Global Tech

Jan 2024–Present

Senior Data Scientist (Machine Learning)

Sunnyvale, CA

Led the launch of project **ANN semantic search** for Walmart international markets by collaborating with cross-functional teams, achieving **6% CTR lift**, **1.4% revenue increase**, and **8% conversion rate improvement**.

- Implemented two-tower architecture with multilingual DistilBERT encoders for query-product semantic retrieval; integrated embeddings with Walmart Shopping Agent (Sparky) for agentic search and context-aware discovery
- Applied supervised fine-tuning (SFT) and reinforcement learning (RLHF) to align semantic embedding model with customer preference
- Built MLOps pipeline from data prep to model retraining and validation to embedding offline indexing, established cross-department **MLOps infrastructure** (Feature Store, Kubeflow pipelines, CI/CD), enabling daily model releases for 20+ Data Scientists in team.

Built **multilingual Typeahead/Autocomplete** ranking system improving conversion by **10 bps** and CTR by **10%**.

- Iterated scoring model from XGBoost to transformer based cross-encoder for improved engagement prediction; optimized for multilingual queries across Walmart international market websites

Engineered a **generative recommender** prototype to address the item cold-start problem based on HSTU algorithm, using RQ-VAE and K-Means clustering to create scalable semantic IDs for over 10M products.

Built a **visual search** engine prototype with **multi-modality** representations (image + text), improving top-1 match accuracy for in-the-wild queries across a multimillion-SKU catalog.

Walmart Global Tech

2020–2022

Data Scientist Intern (Search & Supply Chain)

Sunnyvale, CA

Built **Graph Neural Network ETA model** for online grocery delivery serving 1M+ daily orders; improved delivery time estimates by **5%**.

Developed **last-mile route optimization** system using reinforcement learning; granted *U.S. Patent*: “Methods and apparatus for automatic route assignment.”

RESEARCH EXPERIENCE

Algorithms Research Assistant

Sep 2019–Dec 2023

Virginia Tech Algorithms Lab (NSF-funded)

Blacksburg, VA

Researched scalable optimal transport algorithms with applications to generative AI and distribution alignment; developed novel partial Wasserstein metrics for robust distribution comparison and PU-learning without prior; designed the first graph based parallel algorithms to compute approximate Wasserstein distance (not Sinkhorn distance) on GPU.

Selected Publications (alphabetical authorship)

“A New Robust Partial p -Wasserstein-Based Metric for Comparing Distributions.” ICML 2024.

“A Combinatorial Algorithm for Approximating Optimal Transport in Parallel and MPC.” NeurIPS 2023.

“Computing all Optimal Partial Transports.” ICLR 2023.

EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA

Ph.D. in Computer Science, GPA: 3.9/4.0

Aug 2019–Dec 2024